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Project Title:	Studies of Atomic Free Radicals in a Cryogenic Environment		
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Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
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Space Biology Element:	None		
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Final Report

A wide range of techniques have been applied to the study of Impurity helium Solids, which consist of impurity atoms or molecules or clusters of these impurities surrounded by thin layers of solid helium. These solids are porous and the pores can be filled with superfluid 4He. Ultrasound, X ray diffraction, nuclear magnetic resonance and electron spin resonance have all been applied to determine the nature of these substances. Studies of low temperature chemical reactions, studies of potential quantum fluid behavior and studies to maximize energy storage are have been completed.

Impurity-Helium (Im-He) solids are truly fascinating substances. An important goal of this research program is to investigate these substance by a variety of experimental techniques in order to advance our knowledge of quantum

Task Description:	solids, low temperature chemical reactions and complex materials consisting of nanoclusters of impurities embedded in a mixture of liquid and solid helium. The criterion for quantum liquid and solid behavior is that the thermal de Broglie wave length is comparable to or greater than the mean interparticle spacing. Of course the liquid helium contained in the pores of these substances is a true quantum fluid. Clusters of hydrogen or deuterium molecules embedded in Im-He solids are quantum solids. On the other hand, the heavier impurities such as nitrogen or krypton behave classically. Atomic hydrogen is found to be quite mobile in impurity-helium solids. If low enough temperatures and high enough densities are achieved, it might be possible to observe Bose-Einstein condensation which would be a truly exciting discovery! The assembly of hydrogen atoms would then constitute a separate new quantum fluid. Chemical reactions such as the recombination of H atoms into hydrogen molecules are spontaneous and do not require thermal activation. Even at temperatures near 1.3 K chemical reactions can proceed. The reaction mechanism involves quantum mechanical tunneling and the reactions are called exchange tunneling reactions. Studies of the kinetics of the reaction involving D atoms and hydrogen molecules yielding HD molecules and H atoms have been performed. This reaction is followed by a slower reaction involving D atoms and HD molecules, yielding H atoms and deuterium molecules. Very high concentrations of H atoms can be achieved via exchange tunneling reactions. Attempts have been made to maximize the yield of H atoms to provide high enough concentrations to get into quantum fluid territory. Electron spin resonance experiments have been employed in this work to determine H atom and D atom concentrations. It is important to obtain further details regarding the structure of impurity-helium solids. To this end, x ray scattering and electron spin resonance methods have been employed. Analysis of the x ray diffraction patterns indicate that the impurity atoms tend to form small (nanoscale) clusters. (Each cluster is thought to be surrounded by a very thin layer of solid helium.) As the sample is warmed above the superfluid 4He transition, the clusters grow. The environment surrounding impurity atoms in the clusters may be obtained by observing the detailed structure of the electron spin resonance signals. Satellite lines on the cw resonance signals indicate that the microwave transitions tend to induce spin flips of hydrogen nuclei on neighboring ortho-hydrogen molecules. Pulsed magnetic resonance studies allow satellite lines corresponding to deuteron spin flips to be observed. Such experiments have been conducted. Experiments have been performed to maximize the concentration of nitrogen atoms in impurity helium solids. The nitrogen atom recombination reaction is one of the most energetic chemical reactions, so an impurity helium solid containing high concentrations of nitrogen atoms would provide a very effective energy storage medium. Finally, for the case of impurity-helium solids with free radicals, the presence of unpaired spins leads to the possibility that magnetic forces can suspend a sample, thereby simulating microgravity conditions. This phenomenon has been observed in the course or our research program. Magnetic fields can also be used to transfer impurity helium solids containing paramagnetic atoms between different locations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Impurity-helium solids provide a unique opportunity to study a wide variety of fundamental phenomena as well as to investigate possible practical applications. Impurities may be varied from the heaviest atoms or molecules which behave classically to the lightest atoms or molecules which can give rise to quantum phenomena. The concentration of impurities may also be varied at will. The Im-He solids are porous solids which can be studied with the pores filled with liquid helium or as free standing solids with empty pores. Studies of impurity-helium solids can give significant insight into the nature of porous media in general. The structures of these unique substances bear close resemblance to the light aerogels. For the case of Im-He solids containing the lightest impurities, it might be possible for the ensemble of impurities to act as a quantum fluid or solid. This situation will arise only if the thermal de Broglie wave length is comparable or greater than the mean interparticle spacing between the impurity atoms. Interesting phase transitions might be observed such as Bose-Einstein condensation of hydrogen atoms. Exchange tunneling reactions have also been studied. These reactions can take place in the low temperature regime where thermally activated chemical reactions can no longer occur. They have been used to achieve very high concentrations of atomic hydrogen which ultimately allow entry into the quantum regime. These tunneling reactions may also provide a unique transport mechanism for hydrogen atoms in solid H2. Matrix isolated active free radicals have been considered for possible energy storage application for many years. The impurity-helium solids provide a medium in which very large concentrations of reactive atoms can be achieved and stored. The chemical reactions corresponding to the recombination of these atoms into molecules are highly exothermic. Future applications to special explosives and rocket fuels may ultimately be possible. Finally, tunneling reactions involving isolated reactive atoms may also occur on interstellar dust particles.
	Task Progress: Final Report 2007 Impurity-Helium (Im-He) solids form when a mixture of an impurity gas (N2, D2 or H2 for example) and helium gas is injected into superfluid helium at 1.5 K. If the mixture is subject to an intense radiofrequency discharge immediately prior to injection, the impurity molecular gas sample is partially dissociated so that the impurity atoms produced become embedded in the resulting solid. Studies of Im-He solids by X ray scattering have shown that these solids are composed of nanoclusters of the impurity atoms and molecules each surrounded by a thin layer of solid helium. These nanoclusters become loosely aggregated into a gel within the superfluid helium. A typical cluster is 90 angstroms (9 nm) in diameter. The crystal structures in these nanoclusters have been determined in the aforementioned X ray scattering experiments. More recently the X ray results have been reinterpreted to obtain more realistic models of the crystal structure of the nanocrystallites. Electron spin resonance (ESR) experiments have been underway for several years at Cornell. A continuous wave (CW) X band (9 GHz) ESR spectrometer has been employed to study the concentration and distribution of the atoms (atomic free radicals) embedded in the Im-He clusters. Numerous results obtained with this spectrometer have been reported previously. During the present reporting period, exciting new observations have been made on atomic nitrogen free radicals embedded in molecular nitrogen clusters contained in Im-He solid samples. In recent studies, Im-He solids containing very high (of order E 21 per cubic centimeter) local concentrations of atomic nitrogen have been

Task Progress:

investigated. These atoms were sufficiently closely spaced (about 10 angstroms on average) for very strong interatomic electronic dipolar interactions to be present. In standard ESR experiments, the electronic spins (and associated magnetic moments) flip independently of one another. This is the most probable course of events and is called an allowed transition. For the strong dipolar coupling in concentrated atomic nitrogen samples, it is also possible to observe a so-called forbidden transition in which two spins flip simultaneously in response to a single microwave photon. These so-called spin pair radicals have now been observed in the highly concentrated atomic nitrogen free radicals embedded in the nitrogen clusters contained in the Im-He samples. The effect has been seen in samples containing the most plentiful isotope, nitrogen 14 and those containing the rare isotope, nitrogen 15. The double spin flips were observed by applying one half the steady magnetic field required for observing ESR at 9 GHz. The combined energy of the double (spin pair radical) transition, being proportional to the applied steady field, then gave a resonant frequency matching the 9 GHz spectrometer operating frequency. The spin pair radical signal strength was about three orders of magnitude weaker than that of the main allowed transition. This is understandable in view of the fact that the transition probability of the allowed transition is about three orders of magnitude larger than the transition probability for the (forbidden) spin pair radical transition. The fact that the ratio of the allowed to the forbidden spin pair radical signal is roughly comparable to the ratio of the corresponding transition probabilities leads to the conclusion that a large fraction of the atomic nitrogen sample is participating in the spin pair radical behavior. This in turn suggests that this data along with the atomic nitrogen concentration can be used to estimate the forbidden transition probability and compare with theory. The quality of the spin pair radical data was greatly improved by operating the spectrometer at very high microwave amplitudes. No evidence for saturation effects were seen, so that microwave powers considerably greater that the capabilities of the existing microwave spectrometer will be required to observe saturation.

The recently constructed pulsed ESR spectrometer has now been in operation for a few years. Prior studies via CW ESR spectroscopy of impurity-helium solids containing hydrogen and deuterium atomic free radicals have been effective in determining local and average radical concentrations and in following the progress of exchange tunneling reactions. The CW method suffers, however, from a high degree of inhomogeneous broadening due to nuclei and atomic radicals within the impurity clusters. This broadening obscures some of the weak interactions between the atomic radicals and the nuclei of neighboring molecules. For example, CW ESR is able to resolve satellite lines associated with spin flips of protons on adjacent HD and ortho H2 molecules because the proton magnetic moment is quite large. On the other hand, the satellites associated with the deuteron moments in HD and deuterium molecules cannot be observed due to the small deuteron magnet moment, which is a factor of seven smaller than that of the proton. The standard pulsed ESR methods of echo refocusing (spin echo) allow the study of the interaction of molecular deuterons with the unpaired electrons associated with either the hydrogen or deuterium atomic free radicals. Thus the pulsed ESR measurements can provide complementary information about the HD and molecular deuterium neighbors of H and D atoms.

The ESEEM (electron spin echo envelope modulation) method is the preferred technique for extracting information about the deuterons associated with the neighboring HD and deuterium molecules. The closest molecular neighbors for an ensemble of H atoms or D atoms can be monitored via the ESEEM technique. The technique involves a sequence of spin echo experiments where the value of the temporal spacing tau between the 90 and 180-degree pulses is varied for successive trials. The echo amplitudes are plotted vs. the value of tau. The resultant spin echo amplitude for successive trials displays a periodically varying echo envelope or echo modulation. The depth of modulation is an extremely important quantity, which provides a measure of the number of HD and molecular deuterium near neighbors. An analysis for pure deuterium samples shows that approximately half of the deuterium atoms live on the surfaces of the nanoclusters. This is in accord with a recent theory.

It was found in earlier CW ESR studies in this laboratory that the satellite lines associated with molecular proton spin flips were unchanged around the hydrogen atomic free radicals as the tunneling exchange reactions involving deuterium atoms proceeded, thus reducing the HD population and producing deuterium molecules. For this case the satellite intensities remained unchanged indicating very little change took place in the populations of neighboring molecules. In contrast to this behavior, the satellites on the deuterium atomic free radicals associated with proton spin flips on neighboring molecules were found to decrease in intensity as the exchange tunneling reaction progressed. The ESEEM studies have provided further information on the population of molecules surrounding the atomic free radicals during the course of exchange tunneling reactions in the mixed hydrogen-deuterium systems.

The recent ESEEM studies were in good agreement with the CW data described in the previous paragraph. It is assumed that the bulk and surface populations of atoms for a cluster are approximately equal as was determined for the case of the pure deuterium Im-He solids discussed earlier. By fitting the ESEEM data echo modulation at the beginning of a run and comparing it with data taken six hours later, a significant increase in the population of deuterium molecules and a concomitant decrease in that of HD molecules occurred during the course of the run. This is consistent with the idea that deuterium atoms slowly ate away at the HD molecule population to form deuterium molecules.

Further results have been obtained via X ray scattering from Im-He samples at the National Synchrotron Light Source (NSLS) at the Brookhaven National Laboratory. Samples containing purely molecular oxygen impurities have displayed interesting behavior when subjected to high magnetic fields. Preliminary observations showed a large diffraction peak appearing in the X ray spectrum upon applying magnetic fields of 0.5 Tesla and 3.0 Tesla respectively. The peak was considerably larger at the higher field but was completely absent for zero fields. It is well known that oxygen molecules are highly magnetic as manifested, for example, by the large deflections of streams of liquid oxygen being poured through magnetic field gradients. One possible explanation of the observed X ray behavior is the orientation of the molecular magnetic moments in the Im-He clusters as the magnetic field is raised. Further data will be required before this interesting phenomenon can be understood. A recent analysis of recent X ray data has been performed with collaborators in the Institute for Low Temperature Physics and Engineering (Ukraine) and at Rutgers University to determine the crystal structure in Im-He nanoclusters containing Krypton and Argon as the impurities. This data covered a broader range of scattering angles than earlier data. The analysis indicates that the prevailing structures are icosahedral rather than fcc as previously suggested. This work is now continuing.

A collaboration with a group at the University of Turku in Finland led by Dr. Sergey Vasiliev has recently been established. The goal of this research is to study the properties of atomic hydrogen embedded in solid molecular hydrogen at temperatures below 1 K. Electron spin resonance in the 128 GHz frequency range (approximately 2 mm wave length) has been used to study the spins of the H atom free radicals. The embedded atoms are contained in films of molecular hydrogen 150 layers thick. The samples can be cooled to less than 100 mK via a dilution refrigerator. The ESR technique provides a method for monitoring the relative population of the two lowest hyperfine states. The most dramatic result of this research has been the observation of a drastic departure (a factor of three!) from the classical Boltzmann distribution. An exciting possibility is that this may be the signature of quantum statistical (Bose-Einstein)

	correlations in the ensemble of hydrogen atoms. It is known that hydrogen atomic radicals are quite mobile in molecular hydrogen as a result of tunneling exchange reactions. Thus the atoms are not firmly locked to lattice sites. This raises the possibility that wave function overlap can occur between neighboring atomic free radicals, which could provide a mechanism for Bose-Einstein correlations. The results of this research have been published Sept. 1, 2006 in Physical Review Letters. Work is continuing actively in this area at Turku and Cornell. Very preliminary new measurements seem to indicate that very large concentrations (1020 atoms per cubic centimeter) can be obtained in these samples. With further improvement in the technique, significantly larger concentrations and energy storage capabilities may perhaps be obtained.
Bibliography Type:	Description: (Last Updated: 03/20/2008)
Abstracts for Journals and Proceedings	Vasilyev S, Jarvinen J, Khmelenko VV, Lee DM. "Experimental Observation of Atomic Hydrogen Stabilized in Thin Films of Molecular H2 at Temperatures < or = 100 mK." Presented at the 24th International Conference on Low Temperature Physics, Orlando, FL, August 10-17, 2005. LOW TEMPERATURE PHYSICS: 24th International Conference on Low Temperature Physics - LT24. AIP Conference Proceedings 850, 2006. p. 81-82. http://dx.doi.org/10.1063/1.2354614 , Mar-2006
Abstracts for Journals and Proceedings	Bernard EP, Khmelenko VV, Vehmanen E, Borbat PP, Freed JH, Lee DM. "Two-pulse electron spin echo study of deuterium-helium solids." Presented at the 24th International Conference on Low Temperatures Physics, Orlando, FL, August 10-17, 2005. LOW TEMPERATURE PHYSICS: 24th International Conference on Low Temperature Physics - LT24. AIP Conference Proceedings 850, 2006. p. 372-373. http://dx.doi.org/10.1063/1.2354742 , Mar-2006
Abstracts for Journals and Proceedings	Vehmanen E, Khmelenko VV, Bernard EP, Kunttu H, Lee DM. "ESR investigations of spin-pair radicals in nitrogen-helium solids." Presented at the 24th International Conference on Low Temperatures Physics, Orlando, FL, August 10-17, 2005. LOW TEMPERATURE PHYSICS: 24th International Conference on Low Temperature Physics - LT24. AIP Conference Proceedings 850, 2006. p. 374-375. http://dx.doi.org/10.1063/1.2354743 , Mar-2006
Abstracts for Journals and Proceedings	Khmelenko VV, Bernard EP, Vehmanen E, Lee DM. "Pulse and continuous wave Electron Spin Resonance investigations of H and D atoms in impurity-helium solids." Presented at the 24th International Conference on Low Temperature Physics, Orlando, FL, August 10-17, 2005. LOW TEMPERATURE PHYSICS: 24th International Conference on Low Temperature Physics - LT24. AIP Conference Proceedings 850, 2006. p. 376-377. http://dx.doi.org/10.1063/1.2354744 , Mar-2006
Abstracts for Journals and Proceedings	Bernard EP, Khmelenko VV, Vehmanen E, Borbat PP, Freed JH, Lee DM. "Pulse electron spin resonance method for investigation of atoms in impurity-helium solids." Presented at the 24th International Conference on Low Temperature Physics, Orlando, FL, August 10-17, 2005. LOW TEMPERATURE PHYSICS: 24th International Conference on Low Temperature Physics - LT24. AIP Conference Proceedings 850, p. 1659-1660. http://dx.doi.org/10.1063/1.2355345 , Mar-2006
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Kiryukhin V, Kiselev SI, Lee DM. "Impurity-helium solids: Chemistry and physics at 1.5 K." Journal of Low Temperature Physics. 2004 Jan;134(1-2): 133-43. http://dx.doi.org/10.1023/B:JOLT.0000012546.86092.4a , Jan-2004
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Kiryukhin V, Kiselev SI, Lee DM. "ESR and X-ray investigations of deuterium atoms and molecular in impurity-helium solids." Journal of Low Temperature Physics, 2004 Jan;134(1-2): 169-74. http://dx.doi.org/10.1023/B:JOLT.0000012551.39115.10 , Jan-2004
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Lee DM. "Paramagnetic attraction of impurity-helium solids." Journal of Low Temperature Physics. 2004 Jan;134(1-2): 175-80. http://dx.doi.org/10.1023/B:JOLT.0000012552.49280.a0 , Jan-2004
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Lee DM. "Stabilization of high concentrations of nitrogen atoms in impurity-helium solids." Journal of Low Temperature Physics. 2004 Jan;134(1-2): 199-204. http://dx.doi.org/10.1023/B:JOLT.0000012556.69060.eb , Jan-2004
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Kiryukhin V, Kiselev SI, Lee DM. "Deuterium atoms and molecules in nanoclusters of molecular deuterium." Physical Review B. 2004 Mar;69(10):104201. http://dx.doi.org/10.1103/PhysRevB.69.104201 , Mar-2004
Articles in Peer-reviewed Journals	Bernard EP, Boltnev RE, Khmelenko VV, Lee DM. "Long term stability of H atoms in HD-D2-He solids." Journal of Low Temperature Physics. 2005 Feb;138(3-4): 829-34. http://dx.doi.org/10.1007/s10909-005-2310-3 , Feb-2005
Articles in Peer-reviewed Journals	Ahokas J, Jarvinen J, Khmelenko VV, Lee DM, Vasiliev S. "Exotic behavior of hydrogen atoms in solid H2 at temperatures below 1 K." Physical Review Letters. 2006 Sep 1;97(9):095301. PMID: 17026374 , Sep-2006
Articles in Peer-reviewed Journals	Kiryukhin V, Bernard EP, Khmelenko VV, Boltnev RE, Krainyukova NY, Lee DM. "Noble-gas nanoclusters with fivefold symmetry stabilized in superfluid helium." Physical Review Letters 2007 May 11;98(19):195506. PMID: 17677631 http://dx.doi.org/10.1103/PhysRevLett.98.195506 , May-2007
Articles in Peer-reviewed Journals	Khmelenko VV, Kunttu H, Lee DM. "Recent progress in nanostructured impurity helium solids." Journal of Low Temperature Physics 2007 Jul;148(1-2):1-31. Review. http://dx.doi.org/10.1007/s10909-007-9353-6 , Jul-2007
Articles in Peer-reviewed Journals	Khmelenko VV, Bernard EP, Vasiliev S, Lee DM. "Tunnelling chemical reactions of hydrogen isotopes in quantum solids." Russian Chemical Reviews. 2007 Dec;76(12):1107-21. Review. http://dx.doi.org/10.1070/RC2007v076n12ABEH003729 , Dec-2007